1. PROBLEM STATEMENT

Paved surfaces require regular monitoring to identify deterioration before it becomes a safety or maintenance concern. Manual inspections can be labor-intensive and inconsistent, while image-based methods face limitations related to lighting and surface-depth measurement. These challenges complicate the collection of repeatable condition data. The following sections present the need for improved pavement-condition monitoring, the objective of PaveWatch, and relevant background on existing inspection approaches.

1.1 Need Statement

Paved surfaces such as sidewalks, parking areas and campus roads gradually develop cracks, potholes, settlement, and other surface distress because of traffic loading, moisture and drainage conditions [1]. These changes can reduce serviceability and create hazards, but they may be difficult to identify early or compare across inspection dates. An audit by the Los Angeles City Controller found that, over the five fiscal years preceding the audit, the city faced more than 1,700 sidewalk injury claims and 1,020 lawsuits, paying over $35 million in settlements. The City lacked a complete inventory of sidewalk repair needs, while a comprehensive assessment was estimated to cost over $10 million and take two to three years [2]. A 2019 New York City Comptroller audit of the Parks Department's Trees & Sidewalks Program found an average wait of 101 days from a service request to inspection and 419 days from inspection to repair [3]. Current pavement-condition methods include manual surveys and 2D imaging, but these approaches can be labor-intensive, variable between inspections, and affected by environmental conditions or limitations in measuring surface depth [4][5]. Maintenance organizations therefore need a practical method for collecting repeatable, location-referenced condition data and identifying changes that warrant further inspection or repair. This need is directly relevant to Mississippi State University, where Campus Services is responsible for sidewalks, parking facilities, roadways, paving, preventive maintenance, and accessibility improvements [6].

1.2 Objective Statement

The objective of PaveWatch is to provide an unmanned aerial vehicle (UAV) platform for repeatable inspection of paved areas, such as sidewalks and parking lots, and location-referenced reporting of their condition. An operator marks the survey area and flight path, and the UAV collects surface and position data along that path. PaveWatch will combine multiple sensor inputs to address limitations of individual sensing methods. The system will organize the collected data, compare it with an earlier survey when available, and identify candidate changes that warrant closer inspection, such as signs of pavement settlement or heaving. Its report will provide a list of candidate concerns and a preliminary ranking for review. As the final step in reporting, designated municipal or campus maintenance personnel will review the findings and determine which locations require further inspection or repair and how that work should be prioritized. PaveWatch is meant to support maintenance and engineering decisions, not to replace the judgment of a qualified inspector or diagnose structural problems on its own.

1.3 Background and Related Work

Pavement condition depends on both the surface and the supporting soil. Moisture infiltration and inadequate drainage can weaken underlying materials, while expansive soils can swell as they absorb water and cause pavement heave or unevenness [1]. These mechanisms help explain the need for repeated monitoring as environmental conditions change. Existing assessment methods include standardized visual surveys, 2D and 3D imaging, and UAV photogrammetry [5][7][8]. Manual visual surveys are labor-intensive and subject to variability between inspectors. Two-dimensional imaging can document visible distress, but shadows, stains, poor contrast, and image resolution complicate interpretation, and these systems do not directly measure surface depth [5]. Three-dimensional imaging adds depth information through additional sensing equipment and processing [5]. UAV inspection also requires flight planning, safe site operations, and processing and interpretation of collected data [8][9]. Automated collection therefore does not, by itself, eliminate the need to interpret and review inspection results. A formal patent search is still required before making any claim of novelty or patentability.

2. REFERENCES

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